

The Price of Grain

How India prices its harvest — for farmers, for ration shops, for distilleries, for the world market — and why pairing the Minimum Support *Price* with a Minimum Support *Volume* is the design that creates value on both sides of the ledger.

FCI-warehouse-MSPvsMSV · verified-source synthesis · figures carry season vintages

§ 1

One kilogram of rice, six government prices

India does not have *a* price of grain. It has a ladder of administered prices, each one a policy lever, and the gaps between the rungs are where every subsidy, every incentive, and every distortion in the system lives.

₹4,000/qtl*	FCI economic cost of rice HOLDS AT MSP + procurement incidentals + milling + freight + storage + interest. What a quintal actually costs the state by the time it sits in a godown.
₹3,535	Paddy MSP in rice terms BUYS AT KMS 2025-26 common paddy MSP ₹2,369/qtl ÷ 0.67 milling outturn. The farmer-side guarantee, converted to the commodity the state stores.
₹2,800	OMSS reserve price — private trade SELLS AT What a private processor pays at FCI's weekly e-auction (Jan 2025 tiering).

₹2,320	Rice for ethanol distilleries SELLS AT Fixed pan-India rate, ESY 2025-26 (was ₹2,250). Every quintal moves at ~₹1,700 below economic cost — an off-books transfer to the blending programme.
₹2,250	OMSS to states & Bharat-brand channel SELLS AT ₹2,250 for state governments (12 LMT cap), ₹2,400 for NAFED/NCCF milling into Bharat Rice/Atta.
₹300	NFSA central issue price GIVES AT ₹3/kg rice under the Act — and ₹0 under PMGKAY free-grain. The terminal rung: 90% of offtake exits here.

The rice ladder, 2025-26 vintages. *Economic cost is order-of-magnitude (DFPD/FCI cost heads); every other rung is a notified rate. The state buys at rung 2, carries to rung 1, and disposes at rungs 3-6 — the vertical gaps are the food subsidy.

Three features of this ladder define Indian farm pricing:

- **The buy rung is a promise, not a market.** MSP is announced for 23+ crops each season on CACP cost formulas ($A2 + FL + 50\%$), but it binds only where a procurement machine stands behind it — which is rungs, mandis, and godowns, not gazette notifications.
- **The sell rungs are policy dials.** OMSS reserve prices, the ethanol rate, and issue prices move by executive order — sometimes twice a season (rice-for-ethanol: ₹2,800 → ₹2,250 in one January). Government adjusts the *disposal* price to steer private demand the way a central bank adjusts repo.
- **Parallel ladders exist per crop.** Sugarcane runs the same architecture in mirror image: a statutory floor mills must pay (FRP ₹355/qtl, +29% since 2019) against a frozen output price (sugar MSP ₹31/kg since Feb 2019) — a policy-made squeeze that produced ₹16,087 crore of farmer arrears by February 2026.

What the farmer actually receives

The distance between announced price and received price is the central empirical fact of the system. Three measurements, all from verified official series:

Coverage — who sells at MSP at all

Among households that actually *sold* the crop, 13% of paddy sellers and 16% of wheat sellers sold to a procurement agency (NSS 70th round unit data; the famous "6%" figure divides by *all* agricultural households). Official season counts: ~1.19 crore paddy farmers paid in KMS 2024-25; 25.1 lakh wheat farmers in RMS 2025-26. For the other twenty-one MSP crops, coverage is a few percent in the best year and zero in most.

Volume — the share of the harvest the promise reaches

PROCUREMENT AS % OF PRODUCTION · LATEST FULL YEAR PER CROP

CROP	SHARE	AGENCY	READING
Rice	37.1%	FCI + states	standing operation
Wheat	22.6%	FCI + states	standing operation
Ragi	14.8%	states (PDS mandates)	scheme-driven
Mustard	8.8%	NAFED/PSS	crash-year episodes
Jowar / bajra / moong	4–8%	mixed	episodic
Gram, sunflower, groundnut, masur	<2%	NAFED/PSS	token
Tur, urad, soybean, maize	≈0%	—	notification only

Reliability — the variance that breaks sowing decisions

Year-to-year coefficient of variation of procured volume: rice 24%, wheat 21% — versus gram 80%, tur 109%, bajra 110%, maize 120%, urad 143%, soybean 163%. Cotton is the pure case: 124 lakh bales bought in 2019-20, **zero** in 2021-22 and 2022-23, ~100 lakh again in 2024-25. For a farmer choosing what to sow,

a guarantee with 120% variance is not a guarantee; it is a lottery whose ticket is a season's income.

This is the precise sense in which "MSP for 23 crops" is really one guarantee, one semi-guarantee, and twenty-one lotteries — and why the binding reform variable is *volume*, not price.

§ 3

The trade boundary: export and import parity

Domestic administered prices float inside a boundary set by world prices. The rules of thumb per crop class:

- **Rice — export parity above MSP in normal years.** India supplies ~40% of world rice trade; when non-basmati export prices exceed MSP-derived cost, private trade absorbs surplus at no fiscal cost. The 2022–24 export bans and duties *deliberately broke* this parity to hold domestic prices down — the mirror-image intervention to procurement, executed on the sell side of the boundary.
- **Wheat — the marginal case.** Indian wheat is export-viable only in world-price spikes (2021-22: record 433 LMT procurement *and* record exports, followed by the May 2022 ban within a fortnight of a heatwave). MSP sits close to import parity, so both boundaries are policy-active.
- **Pulses and edible oils — import parity below MSP.** Landed tur, urad, masur, yellow peas, and palm/soy oil routinely undercut MSP. A volume guarantee for these crops *without tariff coordination* is fiscally endless: the state would buy at MSP while imports set the market price below it, forever. Duty policy and MSV are one instrument here, not two.

A product-level parity table (APEDA export unit values and DGCI&S import unit values against MSP, by crop and year) is being compiled as a companion dataset — `data/trade_price_parity.csv` in the repository. The qualitative boundaries above are stable; the table adds the per-year magnitudes.

The MSP + MSV ratio: one number, tuned per crop

Define θ (the MSV ratio): the share of a crop's marketed surplus the government commits — in advance, before sowing — to purchase at MSP. Today θ is an accident of infrastructure: ~ 0.4 for rice, ~ 0.3 for wheat, ~ 0 for everything else. The reform is to make θ an explicit, published policy variable.

Why a partial ratio creates outsized farmer value

Farmer revenue per unit: $R = \theta \cdot \text{MSP} + (1-\theta) \cdot P_{\text{mkt}}$

Revenue variance: $\text{Var}(R) = (1-\theta)^2 \cdot \text{Var}(P_{\text{mkt}})$

Variance falls with the *square* of the ratio: a θ of just 0.3 removes 51% of price-revenue variance; $\theta = 0.5$ removes 75%. The first tranche of guaranteed volume does most of the stabilization work — because what destroys farm incomes is the crash-year tail, and a pre-committed volume floor is exactly a put option on that tail. Three further farmer-side effects compound it:

- **Credit:** a government purchase commitment is bankable collateral; kisan-credit pricing against an assured $\theta \cdot \text{MSP}$ cash flow beats pricing against a lottery.
- **Sowing signal:** announced *before* sowing, per-district volume caps steer acreage the way Haryana's ₹7,000/acre diversion payments have failed to — because they change expected revenue, not just add a side payment.
- **Bargaining floor:** even the $(1-\theta)$ share sold privately clears nearer MSP when traders know the state stands ready at the margin — the documented mandi-price support effect of procurement presence.

Why a capped ratio creates government value

Fiscal exposure: $C = \theta \cdot Q \cdot (\text{MSP} + \text{carrying} - \text{disposal price})$ — bounded ex-ante by the cap.

Against: open-ended MSP-for-all $\approx \text{₹}3.4$ lakh crore/yr (+95% on the food subsidy).

- **The exposure is capped and known at budget time** — unlike the sugarcane FRP model, where an uncapped statutory obligation with no viability check manufactured ₹16,087 crore of arrears, or uncapped bajra buying that ended in loss-making resale.
- **The purchased volume is a merchant book, not a write-off.** The state already monetizes stock through four proven channels: OMSS e-auctions to processors (71+ LMT wheat in 2023-24), the ethanol programme (52–72 LMT rice; a tender clause now *mandates* 40% FCI-grain sourcing for distillery supplies), Bharat-brand retail, and PDS/welfare offtake. MSV extends a working disposal machine to new crops — pulses to dal mills, maize to feed and distilleries, millets to PDS/ICDS mandates.
- **Storage physics is respected.** Public capacity (~ 777 LMT covered) caps feasible procurement at roughly 30% of the ~ 261 MT marketed surplus of the major crops. θ per crop is chosen *inside* that envelope — the constraint becomes a design input instead of a crisis.
- **Counter-cyclical release earns the spread.** Buy at MSP in the glut, release through OMSS in the tight year: the state's buffer function, done deliberately, partially self-finances.

The ratio, tuned per crop

RECOMMENDED MSV RATIO (θ) BY CROP CLASS · SYNTHESIS OF VERIFIED FINDINGS

CROP CLASS	θ	DISPOSAL CHANNEL (PRE-CONTRACTED)	RATIONALE
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Rice, wheat	0.30–0.40	PDS/NFSA + OMSS + ethanol	hold, don't extend — stocks already exceed norms 2×
Pulses (tur, urad, masur, gram)	0.20–0.30	dal-mill e-auctions, PDS pilot, buffer	storable; NAFED cost-efficient <i>relative</i> to FCI; 100% tur/urad/masur commitment already legislated to 2028-29 — cap it instead
Maize, millets	0.15–0.25	ethanol (NAFED/NCCF ₹2,291/qtl loop), feed, ICDS/PDS millet mandates	the complete MSV circuit already running — formalize it
Oilseeds	0.10 + PDP	own-account crushing (HAFED model)	import parity below MSP — θ must stay small; pair with duty policy; deficiency payments do the rest
Cotton	0.15 floor	CCI → mills	convert the zero-or-everything cycle into a standing floor
Perishables	0	—	PDP benchmarked on e-NAM clearing prices + contracted processor offtake (Operation Greens' own conclusion); the state cannot hold tomatoes

Three design rules the evidence keeps repeating

1. **Announce $\theta \cdot Q$ before sowing; never after.** Post-harvest volume decisions are politics; pre-sowing volume commitments are policy.
2. **Contract the exit before the entry.** Every quintal procured under MSV has a named disposal channel — a standing e-auction calendar, a blending-mandate allocation, a PDS quota — priced and published. The July 2023 ethanol pause showed what happens to processor trust when the exit is discretionary.
3. **Budget the rung-gap openly.** Selling at ₹2,320 what costs ₹4,000 is a legitimate policy transfer — when it appears as a line item, not as FCI balance-sheet sediment.

For the farmer, MSP+MSV converts an announced price into a bankable quantity: three-quarters of revenue variance removed at $\theta = 0.5$, credit access against a sovereign cash flow, and a sowing-time signal that actually moves acreage.

For the government, it converts an open-ended moral commitment into a budgeted, storage-feasible position with four working monetization channels — at roughly a third of the fiscal cost of a universal legal MSP, with the crash-year tail (the part that causes farm distress and its politics) covered first.

The one-line design: *MSP without volume is a promise; MSV without disposal is a warehouse fire in slow motion. The instrument that works is a pre-announced volume ratio with a pre-sold exit.*

Sources & method. Synthesis of a 102-agent adversarially-verified research run (23/25 claims confirmed 3-0 or 2-1; DFPD Annual Report 2024-25, PIB Year-End Review 2025, IEG WP379/WP420, RAS 2021, data.gov.in Rajya Sabha annexures), the Haryana forward/backward-linkage brief (HAFED, OMSS, BPCL ethanol tenders via ChiniMandi), ChiniMandi sugar-season 2025-26 statistics, and the project's own datasets (procurement, stocks vs norms, offtake by scheme, storage capacity, production-vs-procurement, IISFM Phase-5 depots). Full citations, per-claim verification votes, and every CSV: github.com/herrrickshaw/FCI-warehouse-MSPvsMSV. Trade-parity table (APEDA/DGCI&S) in progress at time of publication. Figures carry season vintages; the ₹4,000 economic-cost rung is order-of-magnitude. This is research synthesis, not investment or policy advice.